

SIMATSENGINEERING
C03 ASSESSMENT TOOL 1

COURSE CODE: CSA1610

Course Name: DWDM

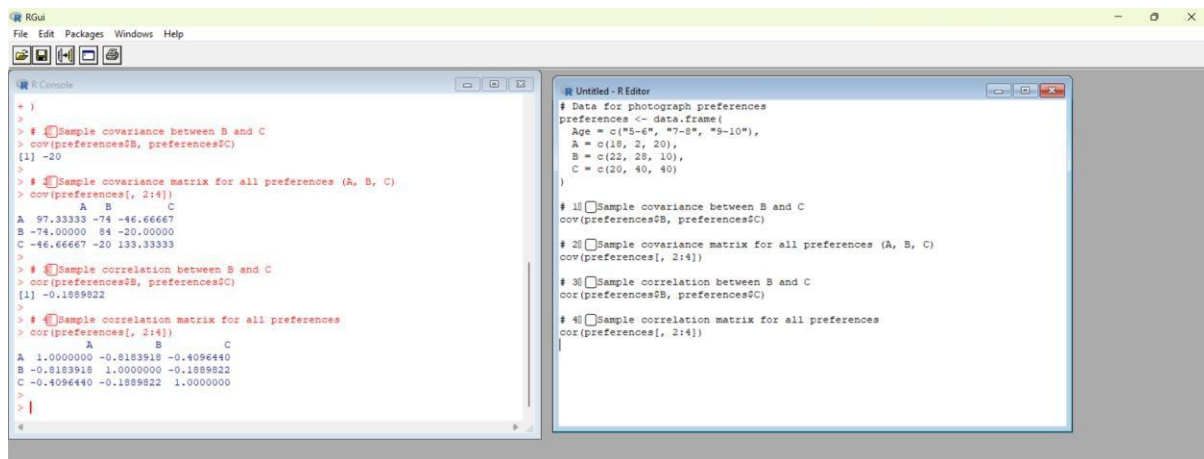
G.Hima Bindu(192511377)

EXPERIMENTS

Q1RCodeforCovarianceandCorrelation:

CODE:

```
#Dataforphotographpreferences preferences<-  
data.frame(  
  Age=c("5-6","7-8","9-10"),  
  A=c(18,2,20),  
  B=c(22,28,10),  
  C=c(20,40,40)  
)  
  
#SamplecovariancebetweenBandC  
cov(preferences$B,preferences$C)  
  
#Samplecovariancematrixforallpreferences(A,B,C)  
cov(preferences[,2:4])  
  
#SamplecorrelationbetweenBandC  
cor(preferences$B,preferences$C)  
  
#Samplecorrelationmatrixforallpreferences  
cor(preferences[,2:4])
```



Q2RCodeforBoxplotandOutlierDetection CODE:

```
#Data:Pointsscoredbyplayers
```

```
points<-c(12,15,14,10,18,20,22,13,11,35,16,17,19,21,23)
```

```
#Createboxplot
```

```
boxplot(points,
```

```
  main="TennisPlayersPointsDistribution", ylab =
```

```
  "Points Scored",
```

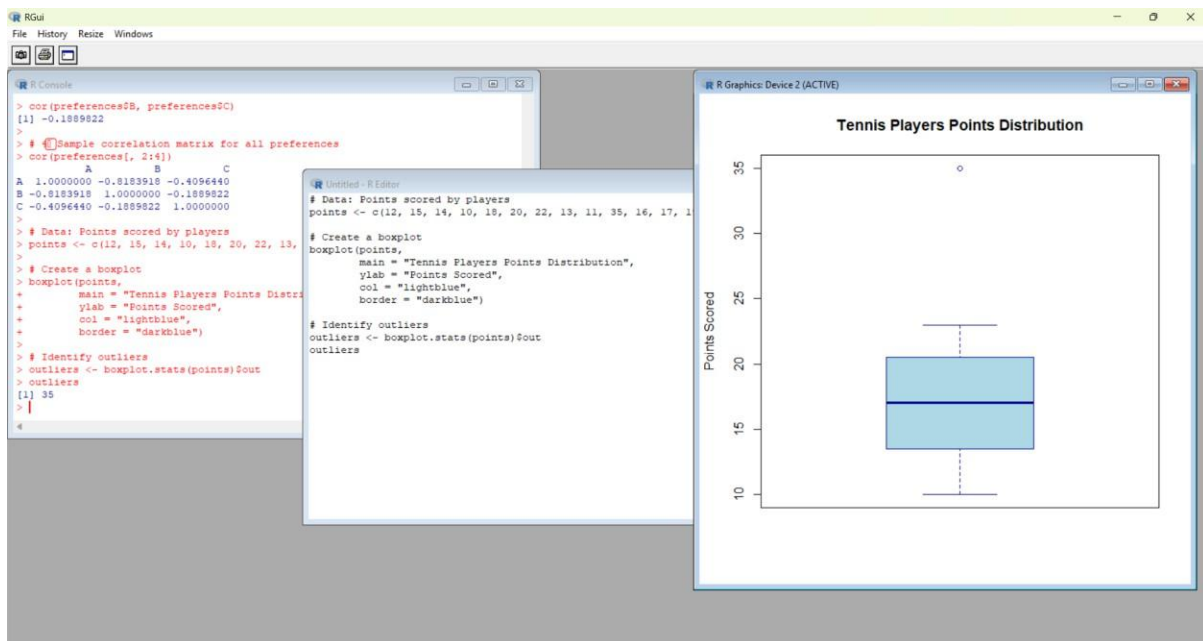
```
  col = "lightblue",
```

```
  border="darkblue")
```

```
#Identifyoutliers
```

```
outliers<-boxplot.stats(points)$out
```

```
outliers
```



Q3RCodeforDecisionTreeClassifier CODE:

#Installandloadrequiredpackages

#install.packages("rpart")

#install.packages("rpart.plot")

library(rpart)

library(rpart.plot)

#Dataset

loan_data<-data.frame(

Age=c("young","young","middle","old","old","old","middle","young","young","old",

"middle","middle","old","young","middle"),

Income=c("high","high","high","medium","low","low","low","medium","low","medium",

"medium","high","medium","medium","low"),

Employment=c("employed","self-

employed","employed","employed","unemployed","self-employed",

"unemployed","employed","unemployed","self-employed","employed","self-employed",

```

    "unemployed","self-employed","employed"),
Credit = c("good","average","good","good","poor","average","poor","average","poor","good",
    "average","good","poor","average","poor"),
Loan=c("yes","yes","yes","yes","no","no","no","yes","no","yes","yes","yes","no","yes","no")
)

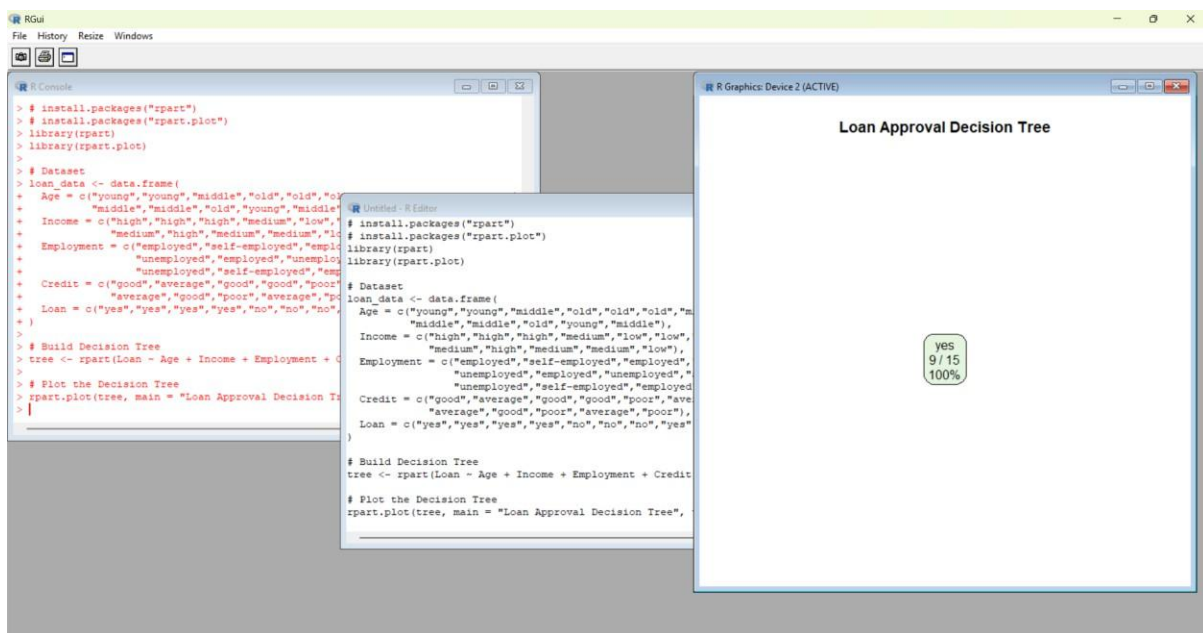
```

```
#BuildDecisionTree
```

```
tree<-rpart(Loan~Age+Income+Employment+Credit,data=loan_data,method= "class")
```

```
#PlottheDecisionTree
```

```
rpart.plot(tree,main="LoanApprovalDecisionTree",type=3,extra=102)
```



Q4RCodeforMean,Median,Range,andBoxplot CODE:

```
#DataforClassAandClassB
```

```
classA<-c(76,35,47,64,95,66,89,36,84)
```

```
classB<-c(51,56,84,60,59,70,63,66,50)
```

```
# Mean, Median, Range meanA
```

```
<- mean(classA) medianA<-
```

```
median(classA) rangeA <-
```

```
range(classA)
```

```
meanB <- mean(classB)
```

```
medianB<-median(classB)
```

```
rangeB <- range(classB)
```

```
#Displayresults
```

```
meanA;medianA;rangeA
```

```
meanB;medianB;rangeB
```

```
#Boxplotcomparison
```

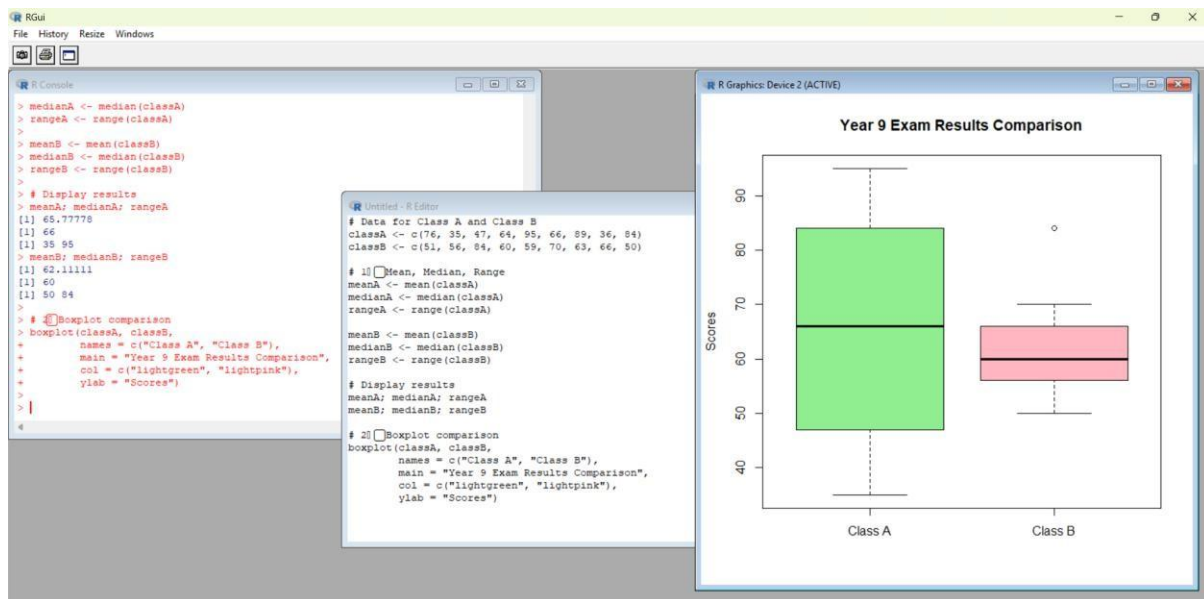
```
boxplot(classA,classB,
```

```
  names=c("ClassA","ClassB"),
```

```
  main="Year9ExamResultsComparison",
```

```
  col=c("lightgreen","lightpink"),
```

```
  ylab="Scores")
```



Q5RCodeforStatisticalAnalysis

CODE:

#Data

```
age<-c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
```

```
fat<-c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,
34.1,32.9,41.2,35.7)
```

#(a)Mean,Median,StandardDeviation

```
mean(age);median(age);sd(age)
```

```
mean(fat);median(fat);sd(fat)
```

#(b)Boxplots

```
par(mfrow=c(1,2))#Displayside-by-side
```

```
boxplot(age,main="BoxplotofAge",col="lightblue",ylab="Age")
```

```
boxplot(fat,main="BoxplotofBodyFat%",col="lightgreen",ylab="%Fat")
```

#(c)ScatterPlotandQ-QPlot

```
par(mfrow=c(1,2))
```

```
plot(age,fat,main="ScatterPlot:AgevsBodyFat",xlab="Age",ylab="%Fat",pch=19, col =
"darkblue")
```

```
qqplot(age,fat,main="Q-QPlot:AgevsBodyFat",xlab="AgeQuantiles",ylab="Body
FatQuantiles",col="purple")
```

